Experiment 9

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Branch: CSE **Section/Group: KRG 3-B**
Semester: 6th **Date of Performance:23/04/2026**
Subject Name: Full Stack Development – II **Subject Code: 23CSH-309**

1. **Aim**: To design and implement a Spring Boot application that integrates third-party APIs including **Stripe Payment Gateway** for handling online transactions and **Gmail SMTP Service** for sending email notifications after successful payments.

2. **Objective**:
   - To understand the concept of third-party API integration
   - To integrate Stripe Payment Gateway for secure online payments
   - To configure and use Gmail SMTP for sending emails
   - To implement REST APIs for payment processing
   - To handle payment success and failure responses
   - To send automated email notifications after successful payment
   - To follow Layered Architecture (Controller – Service – Repository) • To enhance real-world application functionality using external services

3. **Implementation / Code:**

Tools & Technologies Used:-

- Java
- Spring Boot
- Spring Web
- Spring Data JPA
- Maven
- IntelliJ IDEA
- Stripe API
- Gmail SMTP (Java Mail Sender)
- Postman / Browser
- PostgreSQL Database

Implementation Description:-

- A Spring Boot project named Payment Mail Integration is created.

- The project integrates Stripe API for handling online payments securely.

- Users enter card details on Stripe Checkout page and complete payment.

After successful payment, a success page is displayed.

•A confirmation email is automatically sent to the user using Gmail SMTP.

Project Structure:-

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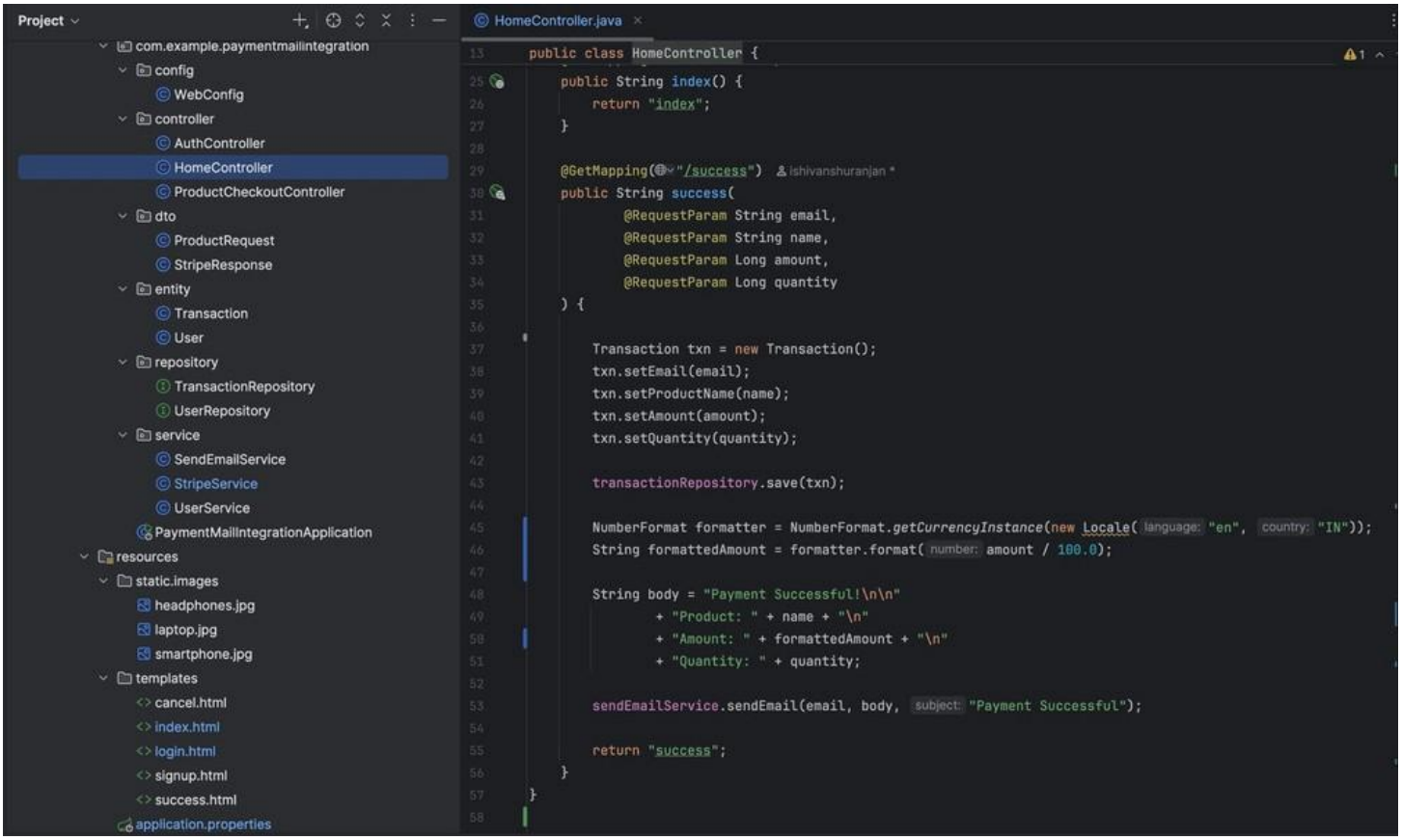
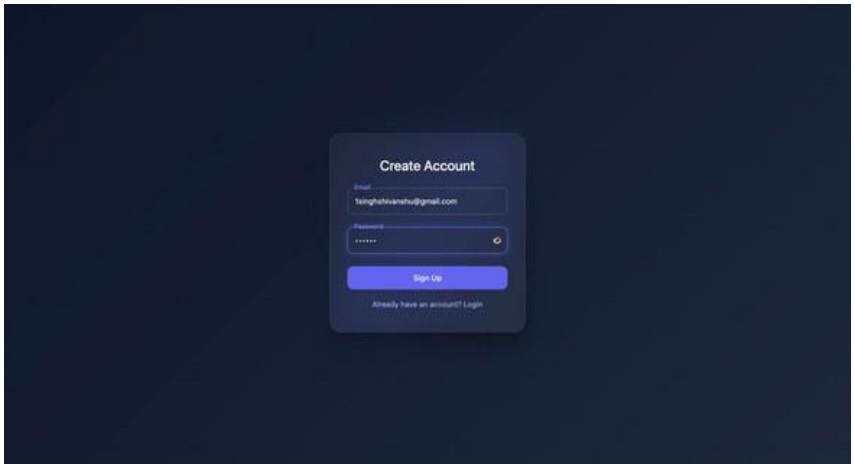


4. Output:

•Spring Boot application started successfully on **localhost:8080**
•Stripe payment page loaded successfully
•Payment completed successfully using test card
•Success page displayed after payment completion
•Email notification sent successfully to registered email
•Email contains product details, amount, and quantity
•Payment flow tested successfully using Stripe test environment
•Application integrated successfully with third-party APIs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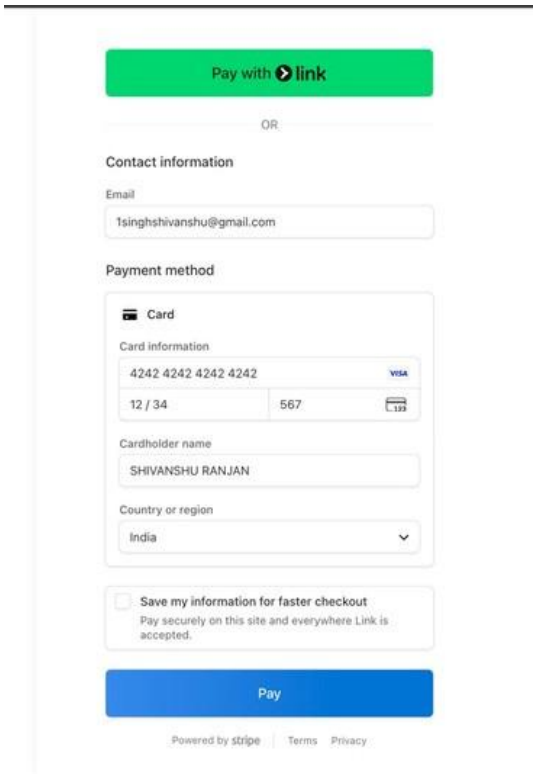
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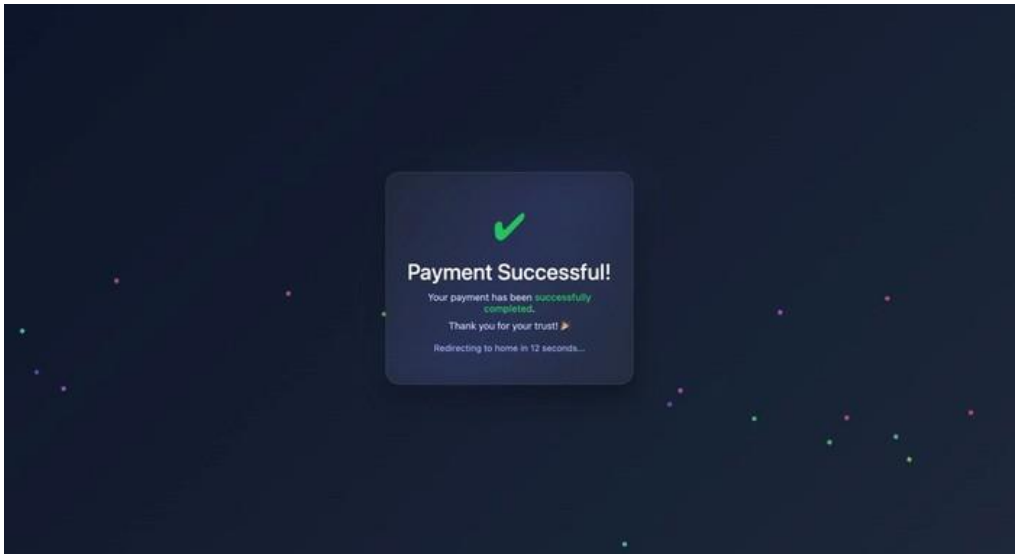

5. Learning Outcomes (What I Have Learnt):

- Understanding of third-party API integration
- Working with Stripe Payment Gateway
- Secure handling of online transactions
- Sending emails using Gmail SMTP
- Implementing real-world payment systems
- Integration of frontend and backend systems
- Usage of REST APIs in real applications
- Handling success and failure scenarios
- Improving application functionality using external services